

RAJIV GANDHI COLLEGE OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF SCIENCE AND HUMANITIES
MATHEMATICS-II ASSIGNMENT-I
PART -A

- 1.State some properties of eigen values.
- 2.Find the sum of eigen values of A^{-1} ,if the matrix $A = \begin{pmatrix} 1 & 3 & 4 \\ 0 & 2 & 5 \\ 0 & 0 & 3 \end{pmatrix}$.
- 3.Find sum and product of eigen values of the matrix $a = \begin{pmatrix} 1 & 2 & 1 \\ 1 & 2 & 0 \\ 1 & 0 & 3 \end{pmatrix}$.
- 4.The product of two eigen values of the matrix $A = \begin{pmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{pmatrix}$ is 16. Find the third eigen value.
- 5.If the eigen values of the matrix $A = \begin{pmatrix} 1 & 1 \\ 3 & -1 \end{pmatrix}$ are 2,-2 then find the eigen values of A^T .
- 6.State Cayley-Hamilton theorem.
- 7.Give two uses of Cayley Hamilton theorem.
- 8.Find the characteristic equation of the matrix $A = \begin{pmatrix} 1 & 2 \\ 0 & 2 \end{pmatrix}$.
- 9.If $\begin{pmatrix} 4 \\ 1 \end{pmatrix}$ is an eigen vector of the matrix $\begin{pmatrix} 5 & 4 \\ 1 & 2 \end{pmatrix}$ find the corresponding eigen value.
- 10.Write the matrix into quadratic form $2x^2 + 8z^2 + 4xy + 10xz - 2yz$.

PART -B

- 11.Find the eigen values and eigen vectors of the matrix $A = \begin{pmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{pmatrix}$.
- 12.Verify Cayley Hamilton theorem for the matrix $A = \begin{pmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & 1 & 2 \end{pmatrix}$ and find A^{-1} and A^4 .
- 13.Diagonalise the matrix $A = \begin{pmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{pmatrix}$.
- 14.Diagonalise the matrix $A = \begin{pmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{pmatrix}$ by orthogonal transformation .
- 15.Reduce $3x^2 + 5y^2 + 3z^2 - 2xy - 2yz + 2zx$ into a canonical form through an orthogonal transformation.